Home Security Audit

Nehaan Merali
30205879
11/11/2025
CPSC 329, Explorations in Information Security and Privacy

Contents

Summary of results ..... -3-
Description of Methods ..... -3-
Table of all devices ..... -7-
Summary of ports ..... -11-
Reflection ..... -11-

Summary of results

In the home network security audit, several Nmap scans were used to identify all active devices and assess potential vulnerabilities within the local network. Starting with a basic ping scan (*nmap -sn*) to locate online hosts, there were 27 online devices and recorded their IP and MAC addresses along with manufacturer details. Subsequent port scans (*nmap -p 1-1000*) revealed that most devices had all ports closed, while a few had open HTTP and HTTPS ports (80 and 443), primarily on routers, smart devices, and set-top boxes. The OS detection scan (*nmap -O*) provided additional insights into each device's operating system and versions, the device purposes and any additional open ports. Attempts to connect to open ports confirmed that most were secure login interfaces or web configuration pages. The audit demonstrated that while the network is well-secured, several devices maintain open web interfaces that could be potential entry points if not properly secured.

Description of Methods

The first terminal command I did was the ***ipconfig -all*** to find the IP address range for my local network.

- The IP address range was 192.168.1.0 - 192.168.1.255.
- This also gave me my personal computers IP address (192.168.1.84)
- All of the devices went throught the exact same nmap scans.

The first NMAP command was ***nmap -sn 192.168.1.0/24***

- This is a basic non-invasive ping scan
- It helped me identify all the hosts on the local network.
- It gave me the IP address, the MAC address, and the manufacture/vendor of the device

```
MAC Address: 8C:6A:8D:84:B7:EF (Vantiva USA)
Nmap scan report for 192.168.1.24
Host is up (0.011s latency).
MAC Address: 00:23:BE:22:86:AF (Cisco Spvtg)
Nmap scan report for 192.168.1.25
Host is up (0.010s latency).
MAC Address: 8C:6A:8D:EC:19:31 (Vantiva USA)
Nmap scan report for 192.168.1.41
Host is up (0.011s latency).
MAC Address: D4:86:60:32:3A:F1 (Arcadyan)
Nmap scan report for 192.168.1.42
Host is up (0.010s latency).
MAC Address: A0:B5:3C:17:21:D5 (Technicolor Delivery Technologies Belgium NV)
Nmap scan report for 192.168.1.65
Host is up (0.014s latency).
MAC Address: 9C:54:DA:0F:94:3E (SkyBell Technologies)
Nmap scan report for 192.168.1.66
Host is up (0.043s latency).
MAC Address: 84:D8:1B:20:B0:9E (TP-Link Technologies)
Nmap scan report for 192.168.1.68
Host is up (0.054s latency).
MAC Address: 44:6F:F8:2B:0E:4E (Dyson Limited)
Nmap scan report for 192.168.1.69
Host is up (0.21s latency).
MAC Address: 3C:31:78:3E:04:23 (Qolsys)
Nmap scan report for 192.168.1.70
Host is up (0.23s latency).
```

```
Nmap scan report for 192.168.1.78
Host is up (0.10s latency).
MAC Address: 64:1C:10:88:B1:F9 (Texas Instruments)
Nmap scan report for 192.168.1.80
Host is up (0.22s latency).
MAC Address: 7E:01:F7:12:38:3C (Unknown)
Nmap scan report for 192.168.1.81
Host is up (0.029s latency).
MAC Address: C6:79:83:C6:69:1B (Unknown)
Nmap scan report for 192.168.1.95
Host is up (0.21s latency).
MAC Address: 84:E6:57:10:39:11 (Sony Interactive Entertainment)
Nmap scan report for 192.168.1.106
Host is up (0.18s latency).
MAC Address: F0:03:8C:C1:5B:66 (AzureWave Technology)
Nmap scan report for 192.168.1.113
Host is up (0.18s latency).
MAC Address: 00:09:B0:1A:3D:69 (Onkyo Technology K.K.)
Nmap scan report for 192.168.1.118
Host is up (0.025s latency).
MAC Address: 88:6B:0F:6C:A6:E4 (Bluegiga Technologies OY)
Nmap scan report for 192.168.1.126
Host is up (0.060s latency).
MAC Address: 10:94:BB:45:E1:99 (Apple)
Nmap scan report for 192.168.1.254
Host is up (0.012s latency).
```

- This is the output of the ***nmap -sn 192.168.1.0/24*** command
- There were a total of 27 hosts online

The second NMAP scan was the port scan, ***nmap -p 1-1000 192.168.1.0/24***

- This identifies any open ports on the devices
- The initial scan was going to take too long to complete, so I scanned each device individually.
  - Each individual scan: ***nmap -p 1-1000 192.168.1.21***
- The result either showed 1000 closed tcp ports or any open ports

```
C:\Windows\System32>nmap -p 1-1000 192.168.1.21
Starting Nmap 7.98 ( https://nmap.org ) at 2025-11-09 21:30 -0700
Nmap scan report for 192.168.1.21
Host is up (0.039s latency).
All 1000 scanned ports on 192.168.1.21 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 24:37:4C:F1:A4:6E (Cisco Spvtg)

Nmap done: 1 IP address (1 host up) scanned in 1.88 seconds

C:\Windows\System32>_
```

- Image shows all ports closed

```
C:\Windows\System32>nmap -p 1-1000 192.168.1.42
Starting Nmap 7.98 ( https://nmap.org ) at 2025-11-11 20:56 -0700
Stats: 0:00:23 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 54.27% done; ETC: 20:57 (0:00:19 remaining)
Stats: 0:00:43 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 93.75% done; ETC: 20:57 (0:00:03 remaining)
Nmap scan report for 192.168.1.42
Host is up (0.0080s latency).
Not shown: 998 closed tcp ports (reset)
PORT    STATE SERVICE
80/tcp  open  http
443/tcp open  https
MAC Address: A0:B5:3C:17:21:D5 (Technicolor Delivery Technologies Belgium NV)

Nmap done: 1 IP address (1 host up) scanned in 47.11 seconds

C:\Windows\System32>_
```

- Image shows two ports open, and the corresponding service of the port

The third and final nmap scan was ***nmap -O 192.168.1.0/24***

- This scan is an attempt to identify the operating system of the device
- It also shows any ports open on the device
- This command also took a long time, so i scanned each device individually, the same process as the port scans.
- This was the scan that gave me the most information.
- It gave me the device type (what is is used for), the Operating System details, and which version of that system it is running.
- It also scanned all ports, not just 1-1000
- From this scan, I was able to identify most of the devices from the Operating System Details section, although there were a few I was unable to identify
- There were also a few times I ran this command and the nmap scan could not find the Operating system and showed an error

```
C:\Windows\System32>nmap -O 192.168.1.21
Starting Nmap 7.98 ( https://nmap.org ) at 2025-11-09 21:31 -0700
Nmap scan report for 192.168.1.21
Host is up (0.019s latency).
Not shown: 998 closed tcp ports (reset)
PORT     STATE SERVICE
8080/tcp open  http-proxy
8086/tcp open  d-s-n
MAC Address: 24:37:4C:F1:A4:6E (Cisco Spvtg)
Device type: media device
Running: Microsoft Windows PocketPC/CE
OS CPE: cpe:/o:microsoft:windows_ce
OS details: Motorola VIP1200-series or Swisscom Bluewin TV digital
set top box (Windows CE 5.0)
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at http
s://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 3.19 seconds

C:\Windows\System32>_
```

- The image shows that the IP scanned:
  - The device type is a media device

- The operating system is Microsoft Windows PocketPC/CE
- The device is a Swisscom Bluewin TV digital set top box, in other words, a cable box for a TV.
- ports 8080 and 8086 are open, and they are http-proxy and d-s-n, respectively
- From the OS details portion, I was able to figure out that this device was a cable box

```
Nmap scan report for 192.168.1.69
Host is up (0.016s latency).
Not shown: 999 closed tcp ports (reset)
PORT     STATE SERVICE
9200/tcp open  wap-wsp
MAC Address: 3C:31:78:3E:04:23 (Qolsys)
No exact OS matches for host (If you know what OS is running on it, see https://nmap.org/submit/ ).
TCP/IP fingerprint:
OS:SCAN(V=7.98%E=4%D=11/9%OT=9200%CT=1%CU=42977%PV=Y%DS=1%DC=D%G=Y%M=3C3178
OS:%TM=6911258E%P=i686-pc-windows-windows)SEQ(SP=102%GCD=1%ISR=10E%TI=Z%TS=
OS:7)SEQ(SP=105%GCD=2%ISR=10A%TI=Z%TS=7)SEQ(SP=106%GCD=1%ISR=109%TI=Z%TS=7)
OS:SEQ(SP=108%GCD=1%ISR=108%TI=Z%TS=7)SEQ(SP=FD%GCD=1%ISR=108%TI=Z%TS=7)OPS
OS:(O1=M5B4ST11NW6%O2=M5B4ST11NW6%O3=M5B4NNT11NW6%O4=M5B4ST11NW6%O5=M5B4ST1
OS:1NW6%O6=M5B4ST11)WIN(W1=FFFF%W2=FFFF%W3=FFFF%W4=FFFF%W5=FFFF%W6=FFFF)ECN
OS:(R=Y%DF=Y%T=40%W=FFFF%O=M5B4NNSNW6%CC=Y%Q=)T1(R=Y%DF=Y%T=40%S=O%A=S+%F=A
OS:S%RD=0%Q=)T2(R=N)T3(R=N)T4(R=N)T5(R=Y%DF=Y%T=40%W=0%S=Z%A=S+%F=AR%O=%RD=
OS:0%Q=)T6(R=N)T7(R=N)U1(R=Y%DF=N%T=40%IPL=164%UN=0%RIPL=G%RID=G%RIPCK=G%RU
OS:CK=G%RUD=G)IE(R=N)

Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 340.92 seconds
```

- Image Error
- NMAP command cannot identify the Operaing System
- I think this is because it is a security system, so it is a more secure system than other devices

For any devices I was unable to identify from the NMAP scans, I went onto my the TELUS SmartHome app.

- This app shows a list of all devices connected to my home network.
- I also gives the device name, their IP address and their MAC address.
- On the app, it showed around 40 devices connected to the network, while my original NMAP scan showed only 27.
- There were also some devices listed on the NMAP scan that were not listed on the TELUS app, and some devices on the TELUS app that did not show up on the NMAP scan

The order of scans I did went more invasive throughout the process, as each scan gave more information about the devices.
There were also many devices that disconnected as I was doing my scans, this was due to devices disconnecting and reconnecting to the network. On the table they are labelled “Disconnected”

Table of all devices

IP Address	MAC Address/ Manufacturer	OS Type (Device type, OS details)	Open Ports	Device	Process
192.168.1.21	24:37:4C:F1:A4:6E Cisco Spvtg	• Media Device • Motorola VIP1200-series or Swisscom Bluewin TV digital set top box (Windows CE 5.0)	8080/tcp, http 8086/tcp, d-s-n	Cable Box	Discovered from OS device type
192.168.1.22	00:25:2E:57:75:02 Cisco Spvtg	• Media device • Motorola VIP1200-series or Swisscom Bluewin TV digital set top box (Windows CE 5.0)	8080/tcp, http-proxy 8086/tcp d-s-n	Cable box	Discovered from OS device type
192.168.1.23	8C:6A:8D:84:B7:EF Vantiva USA	• General purpose • Linux 3.2-4.14	8080/tcp, http-proxy 8086/tcp d-s-n 9080tcp/ glrpc	Network Switch	Discovered from Telus app
192.168.1.24	00:23:BE:22:86:AF Cisco Spvtg	• Media device • Motorola VIP1200-series or Swisscom Bluewin TV digital set top box (Windows CE 5.0)	8080/tcp, http-proxy 8086/tcp d-s-n	Cable Box	Discovered based off other Cisco Spvtg devices
192.168.1.25	C:6A:8D:EC:19:31 Vantiva USA	• General purpose • Linux 3.2-4.14	8080/tcp, http-proxy 8086/tcp d-s-n	Device not found on Telus app or NMAP scans	N/A
192.168.1.41	D4:86:60:32:3A:F1 Arcadyan	• General purpose • Linux 3.2-4.14	80/tcp, http	Device not found on Telus app or NMAP scans	N/A
192.168.1.42	A0:B5:3C:17:21:D5 Technicolor Delivery Technologies Belgium NV	• general purpose • OpenWrt 19.07 (Linux 4.14)	80/tcp, http 443/tcp, https 8080/tcp. http-proxy	Device not found on Telus app or NMAP scans	N/A

192.168.1.65	9C:54:DA:0F:94:3E SkyBell Technologies	• proxyserver|firewall|remote management • Citrix access gateway VPN gateway, Juniper SA2500 SSL VPN gateway or Digi Passport console server	6861/tcp(closed) bittorrent-tracker 6969/tcp acmsoda	Back Door Bell	Discovered by it being the only device in the house from Sky Bell Technologies
192.168.1.66	84:D8:1B:20:B0:9E TP-Link Technologies	• Specialized • TP Link Hs200 smart switch (lwIP 2.0 or later)	999/tcp, abyss	Network Switch	Discovered from Telus app
192.168.1.68	44:6F:F8:2B:0E:4E Dyson Limited	• specialized • Chamberlain myQ garage door opener	All ports closed	Dyson Purifyer heater and fan	Only Dyson device in the house
192.168.1.69	3C:31:78:3E:04:23 Qolsys	Error, Look at image in Description Methods Section	9200/tcp, wap-wsp	Alarm System	Discovered from Telus App
192.168.1.70	12:41:B5:67:10:7E Unknown	• phone • Apple iOS 15.0-15.6 (Darwin 21.1.0-21.6.0)	62078/tcp, iphone-sync	iphone	Discovered from port name (iphone-sync) and OS device type
192.168.1.72	B8:3A:9D:2D:AE:CE Alarm.com	• general purpose • Linux 3.2-4.14	554/tcp, rtsp	House Security System	I knew it Had to do something with security system from Manufacture type. but used the Telus app to figure out which system
192.168.1.73	44:09:B8:AB:8E:F7 Salcomp (Shenzhen)	• Media device|general purpose|phone	8008/tcp, http 8009/tcp, ajp13	Google Chrome Cast	Discovered from Telus app

		• Android 4.1.1, Linux 3.2-4.14, Android 4.2.2(Linux3.4)	8443/tcp https-alt 9000/tcp cslistener 10001/tcp scp-config		
192.168.1.75	46:E7:BF:69:60:54 Unknown	N/A	N/A	N/A	Host went offline
192.168.1.76	2C:71:FF:D8:6E:24 Amazon Technologies	• media device • Android 9(Linux 4.9)	5555/tcp, freeciv 8009/ajp13	Fire Stick	Only device we got from Amazon technologies
192.168.1.77	74:58:F3:EE:82:E4 Amazon Technologies	• media device, phone • Amazon Fire TV or Kindle Paperwhite, CyanogenMod 12 (Android 5.0.2	8009/tcp, ajp13	Fire Stick	Discovered from OS details (amazon fire TV)
192.168.1.78	64:1C:10:88:B1:F9 Texas Instruments	• general purpose| router • Linux 4.15-5.19, OpenWrt 21.02 (Linux 5.4), MikroTik Router OS	53/tcp, domain 111/tcp(filtered), rpcbind 443/tcp https	Thermostat	Discovered from Telus app
192.168.1.80	7E:01:F7:12:38:3C Unknown	N/A	N/A	N/A	Device disconnected
192.168.1.81	C6:79:83:C6:69:1B Unknown	• general purpose • Apple macOS 11(Big Sur) - 13 (Ventura) or iOS 16 (Darwin 20.6.0 - 22.4.0)	49152/tcp, unknown 62078/tcp, iphone-sync	ipad	Knew it was an apple device from the OS details, and the does not say iphone type
192.168.1.84	N/A	N/A	135/tcp, msrpc 139/tcp netbios -ssn 445/tcp, microsoft-ds 8090/tcp, opsmessaging	My Computer	New it was my device from initial IP config command

192.168.1.95	84:E6:57:10:39:11 Sony Interactive Entertainment	• general purpose|game console • FreeBSD 11.X|12.X, Sony embedded	All ports closed	PS5	Only device in the house from sony. also says in OS details
192.168.1.106	F0:03:8C:C1:5B:66 AzureWave Technology	Error . No OS matches	80/tcp, http 443/tcp http 515/tcp, printer 631.tcp, ipp	Canon Printer	Discovered based on port name (515/tcp, printer)
192.168.1.113	00:09:B0:1A:3D:69 Onkyo Technology K.K.	Error . No OS matches	80/tcp, http 5000/tcp, upnp 8008/tcp, http 8009/tcp, ajp13 8080/tcp https-alt 8888/tcp, sun-answerbook 9000/tcp, cslistener 10001/tcp scp-config	TV receiver	Discovered from telus app. It maes sense since it has a lot of ports open to receive data
192.168.1.118	88:6B:0F:6C:A6:E4 Bluegiga Technologies OY	No OS matches, but warning occurred:	All ports closed	Front Door Camera	Discovered from Telus app
192.168.1.126	10:94:BB:45:E1:99 Apple	• phone|general purpose • Apple macOS 10.13 (High Sierra) - 10.15 (Catalina) or iOS 11.0 - 14.3 (Darwin 17.0.0 - 20.2.0)	62078/tcp, iphone sync	iPhone	Discovered from port name (iphone-sync) and OS device type
192.168.1.254	A8:A2:37:14:88:8B Arcadyan	• general Purpose • Linux 3.10-4.11	53/tcp, domain 80/tcp, http 443/tcp https	Device not found on Telus app or NMAP scans	N/A

Summary of ports

After this scan, I tried to connect to some of the open ports (80 [http] or 443[https]). This was The most invasive process.

- Since there were no ssh ports open, I could not ssh into anything.
- For the http ports open, I opened a web browser and typed *http://192.168.1.42*
  - This led me to the Telus Login page, where you needed to enter your Username and Password. I didn't know the correct info, so i went on to the next step
- I also used chatgpt[2] and a website[1] to find any command line arguments to connect to http port
  - curl http://192.168.1.10
  - This printed out the webpages HTML code directly into a terminal
  - Other commands like "nc <IP address> 80" or "telnet <IP address> 80" required external system downloads

I was not able to connect to any of the open ports. I learned that my network is reasonably secure: when I tried to open the device web pages by IP address, the interfaces required the ISP/router account credentials, so an attacker would need the network owner's login (or physical access) to reach those management pages and tamper with devices.

Reflection

One of the most interesting things that stood out to me during this exercise was how long some of the NMAP commands took to complete for different devices. For example, scanning my home alarm system took noticeably longer than scanning an iPhone or a cable box. This made me realize how the complexity and configuration of a device can influence how it responds to network scans. Devices like alarm systems and routers are designed with additional security layers, which can cause delays or partial responses to probes. In contrast, simpler or more open devices like phones or streaming boxes respond faster because their network services are easier to detect
I was also surprised by which devices actually require a Wi-Fi connection. Some devices that I thought were standalone, like an air purifier, depend heavily on internet access for basic functionality. This showed just how interconnected and dependent modern households have become on Wi-Fi, and how each connected device increases the network's potential attack surface. From a security perspective, one concern is that some of the devices have open HTTP or HTTPS ports for remote management. While these interfaces are convenient, they could be exploited if a device isn't regularly updated or uses default credentials. Overall, this exercise was both helpful and eye-opening. It gave me a much clearer understanding of how many devices are actively communicating within my home network and how to figure out which device is which. It showed me once you are connected to someones internet, you can see how many devices are on there, and even determine what each device is

1. https://www.stationx.net/nmap-cheat-sheet/
2. chatGPT